[1]

$$x_i' = \sum_j a_{ij} x_j$$

$$T'_{rs} = \sum_i \sum_j a_{ri} a_{sj} T_{ij}$$

$$T' = A T A^{T_{ar}} = A T A^{-1}$$

$$|T'| = |A||T||A^{-1}| = |T|$$

$\Sigma \leftrightarrow \Sigma'$

$x \leftrightarrow x'$

$\updownarrow \quad \updownarrow$

$y \leftrightarrow y'$

$y = Tx \qquad y' = T'x'$

$x' = Ax \qquad y' = Ay$

$AT = T'A$

$$T' = A T A^{-1}$$

$$T'_{ij} = a_{ik} a_{jl} T_{kl} \leftarrow (A^t = A^{-1})$$

$$A A^t = I$$

$x_i' = a_{ij} x_j$

$y_i' = a_{ij} y_j$

$y_i = T_{ij} x_j$

$y_i' = T_{ij}' x_j'$

$$A = (e_1\ e_2\ e_3)$$

$$A^t = \begin{pmatrix} e_1^t \\ e_2^t \\ e_3^t \end{pmatrix}$$

$$A A^t = \begin{pmatrix} |e_1|^2 & 0 & \\ & |e_2|^2 & \\ 0 & & |e_3|^2 \end{pmatrix}$$

y_i'

$a_{ij} y_j \qquad T_{ij} x_j'$

$a_{ij} T_{jk} x_k$

直交基底 e_k

[2] $\mathbf{a} = a_k e_k$, $\mathbf{b} = b_k e_k$

$$(e_i \times e_j)_k = \epsilon_{ijk}$$

$$(\mathbf{a} \times \mathbf{b})_k = a_i b_j \epsilon_{ijk}$$

$$(\nabla \times \mathbf{v})_k = \partial_i v_j \epsilon_{ijk}$$

[3] 直交基底 e_k

$$(e_i \times e_j)_k = (e_i \times e_j)\cdot e_k = \det(e_i\ e_j\ e_k) = \epsilon_{ijk}$$

(1) $\epsilon_{ijk}\epsilon_{lmk} = \sum_k \det^{\mathrm{T}}(e_i\ e_j\ e_k)\cdot\det(e_l\ e_m\ e_k)$

4階テンソル

$= a\,\delta_{ij}\delta_{lm} + b\,\delta_{il}\delta_{jm} + c\,\delta_{im}\delta_{jl}$ $\quad a=0,\ b=1,\ c=-1$

$$= \sum_k \det\begin{pmatrix} \delta_{il} & \delta_{im} & \delta_{ik} \\ \delta_{jl} & \delta_{jm} & \delta_{jk} \\ \delta_{kl} & \delta_{km} & \delta_{kk} \end{pmatrix}$$

$$= \sum_k \delta_{il}\delta_{jm}\delta_{kk} + \delta_{im}\underbrace{\delta_{jk}\delta_{lk}}_{\delta_{jl}} + \delta_{ik}\delta_{jl}\delta_{mk} - \delta_{ik}\delta_{jm}\delta_{kl} - \delta_{il}\delta_{jk}\delta_{mk} - \delta_{im}\delta_{jl}\delta_{kk}$$

$$= 3(\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}) + 2\delta_{im}\delta_{jl} - 2\delta_{il}\delta_{jm}$$

$$= \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}$$

(2) $\epsilon_{ijk}\epsilon_{ljk} = \sum_j \delta_{il}\delta_{jj} - \delta_{ij}\delta_{jl} = 3\delta_{il} - \delta_{il} = 2\delta_{il}$

2階

$= c\,\delta_{il}$ $\quad c=2$

(3) $\epsilon_{ijk}\epsilon_{ijk} = \sum_i 2\delta_{ii} = 6$

[4]

(1) $\nabla \times (\nabla \times A) = \nabla(\nabla \cdot A) - \nabla^2 A$ $\qquad A \times (B \times C) = (A \cdot C)B - (A \cdot B)C$

(2) $\underbrace{(A \cdot \nabla)B}_{\nabla(A \cdot B) - (A \cdot \nabla)B} + \underbrace{(B \cdot \nabla)A}_{\nabla(B \cdot A) - (B \cdot \nabla)A} + \underbrace{A \times (\nabla \times B)}_{\nabla(A \cdot B) - (A \cdot \nabla)B} + \underbrace{B \times (\nabla \times A)}_{\nabla(B \cdot A) - (B \cdot \nabla)A}$

$= 2\nabla(A \cdot B)$

(3) $\nabla \cdot (A \times B)$

[5]

(1) $\nabla r = \nabla \sqrt{x^2+y^2+z^2} = \frac{1}{2\sqrt{x^2+y^2+z^2}} \begin{pmatrix} 2x \\ 2y \\ 2z \end{pmatrix} = \frac{\vec{r}}{r}$

(2) $\nabla \cdot \vec{r} = 3$

(3) $\nabla \times \vec{r} = \begin{pmatrix} \frac{\partial z}{\partial y} - \frac{\partial y}{\partial z} \\ \frac{\partial x}{\partial z} - \frac{\partial z}{\partial x} \\ \frac{\partial y}{\partial x} - \frac{\partial x}{\partial y} \end{pmatrix} = \vec{0}$

(4) $\nabla f(r) = \frac{\partial f}{\partial r} \nabla r = \frac{df}{dr} \cdot \frac{\vec{r}}{r}$

(5) $\nabla \cdot (f(r)\vec{r}) = \{\nabla f(r)\} \cdot \vec{r} + f(r) \nabla \cdot \vec{r} = \frac{df}{dr} |\vec{r}| + 3 f(r)$

(6) $\nabla \times (f(r)\vec{r}) = 0$

(7) $\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right)$

(8) $\nabla \cdot (A \times (r \times A))$

$$A \times (B \times C) = B(A \cdot C) - C(A \cdot B)$$

$$\nabla \cdot (\nabla T) = (\nabla \cdot \nabla) T = \nabla^2 T$$

$$\nabla \times (\nabla T) = (\nabla \times \nabla) T = 0$$

$$\nabla (\nabla \cdot \mathbb{h})$$

$$\nabla \cdot (\nabla \times \mathbb{h}) = 0$$

$$\nabla \times (\nabla \times \mathbb{h}) = \nabla (\nabla \cdot \mathbb{h}) - (\nabla \cdot \nabla) \mathbb{h}$$

$$\nabla \times A = 0 \iff A = \nabla \varphi$$

$$\nabla \cdot D = 0 \iff D = \nabla \times C$$

[8]

(1) $J_{ij} = \dfrac{\partial x_i}{\partial \xi_j}$

$dx^i = J_{ij}\, d\xi^j$

$\sum_i dx^i dx = \sum_i J_{ij} J_{ik}\, d\xi^j d\xi^k$

$g_{jk} = J_{ij} J_{ik}$

$= J^T J$

$g^{ij} = (J^T)^{-1} J$

$\dfrac{\partial \phi}{\partial x^i} = J^{-1} \dfrac{\partial \phi}{\partial \xi_i}$

$\nabla\phi \cdot \nabla x = \dfrac{\partial \xi^j}{\partial x_i} \dfrac{\partial \xi^k}{\partial x_k} \dfrac{\partial \phi}{\partial \xi^j} \dfrac{\partial x}{\partial \xi^k}$

$= g^{jk} \dfrac{\partial \phi}{\partial \xi^j} \dfrac{\partial x}{\partial \xi^k}$

$d^3r = |\det J| \cdot d^3\xi = \sqrt{g}\, d^3\xi$

$\det J = \pm \sqrt{g}$

↑ 右手系にとる

(2) $\int g^{ij} \dfrac{\partial \phi}{\partial \xi^i} \sqrt{g}\, \dfrac{\partial x}{\partial \xi^j} d^3\xi = -\int x \dfrac{\partial}{\partial \xi^j}\left(g^{ij} \dfrac{\partial \phi}{\partial \xi^i} \sqrt{g} \right) d^3\xi = -\int x \nabla^2 \phi \sqrt{g}\, d^3\xi$ (※)

$\sqrt{g}\, \nabla^2 \phi = \dfrac{\partial}{\partial \xi^i}\left(g^{ij} \dfrac{\partial \phi}{\partial \xi^i} \sqrt{g} \right)$

(3) 球座標

$g_{ij} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2 \sin^2\theta \end{pmatrix}$

$\sqrt{g} = r^2 \sin\theta$

円柱座標

$g_{ij} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

9. 円柱座標 : $\nabla^2 = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2}{\partial \varphi^2} + \frac{\partial^2}{\partial z^2}$

$$\frac{\frac{d}{dr}\left(r\frac{dR}{dr}\right)}{rR} + \frac{1}{r^2}\frac{\Phi''}{\Phi} + \frac{Z''}{Z} = -k^2$$

$$\begin{cases} Z'' = c^2 Z \\ \dfrac{r\frac{d}{dr}\left(r\frac{dR}{dr}\right)}{R} + \dfrac{\Phi''}{\Phi} + (\alpha + k^2) r^2 = 0 \end{cases} \quad \begin{cases} \Phi'' = -n^2 \Phi \\ R'' + \frac{1}{r}R' + \left(c^2 + k^2 - \frac{n^2}{r^2}\right) R = 0 \end{cases}$$

Bessel function

$$y'' + \frac{1}{x}y' + \left(1 - \frac{n^2}{x^2}\right) y = 0$$

$$\rightarrow \quad J_n(x) = \frac{1}{n!}\left(\frac{x}{2}\right)^n - \frac{1}{(n+1)!}\left(\frac{x}{2}\right)^{n+2} + \cdots + \frac{(-1)^k}{k!\,(n+k)!}\left(\frac{x}{2}\right)^{n+2k} + \cdots$$

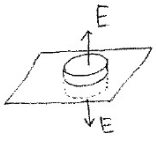
9. $(\nabla^2 + k^2)\phi = 0$

球座標：$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin\theta}\frac{\partial^2}{\partial\varphi^2}$

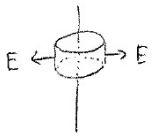
$$r^2\frac{\nabla^2\phi}{\phi} = \frac{\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)}{R} + \frac{\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)}{\sin\theta\,\Theta} + \frac{1}{\sin\theta}\frac{\frac{d^2\Phi}{d\varphi^2}}{\Phi} = -k^2r^2$$

$$\begin{cases} \frac{d}{dr}\left(r^2\frac{dR}{dr}\right) = -k^2r^2R + n^2R \rightarrow & R'' + \frac{2}{r}R' + \left(k^2 - \frac{n^2}{r^2}\right)R = 0 \\ \frac{\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)}{\Theta} + \frac{\frac{d^2\Phi}{d\varphi^2}}{\Phi} = -n^2 \rightarrow & \begin{cases} \frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) = C\Theta \\ \frac{d^2\Phi}{d\varphi^2} = -C\Phi \end{cases} \end{cases}$$

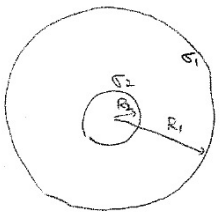
10.

(1) 

$$\epsilon 2ES = \sigma S \rightarrow E = \frac{\sigma}{2\epsilon}$$

(2) 

$$\epsilon E \cdot 2\pi r l = \lambda l \rightarrow E = \frac{\lambda}{2\pi\epsilon r}$$

(3) 

$0 < r < R_2$; $E = 0$

$R_2 < r < R_1$; $E = \frac{\sigma_2 R_2^2}{\epsilon r^2}$

$R_1 < r$; $E = \frac{\sigma_1 R_1^2 + \sigma_2 R_2^2}{\epsilon r^2}$

$$\int_{-\frac{a}{2}}^{\frac{a}{2}} \frac{dz'}{\sqrt{r^2+(z-z')^2}} = \ln \frac{\left(\sqrt{z^2-2az+r^2+\frac{a^2}{4}} - z + \frac{a}{2}\right)}{\left(\sqrt{z^2+2az+r^2+\frac{a^2}{4}} - z - \frac{a}{2}\right)}$$

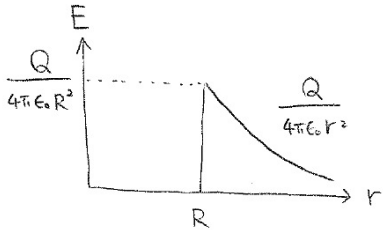
$$= \ln \frac{\sqrt{(z-\frac{a}{2})^2+r^2} - (z-\frac{a}{2})}{\sqrt{(z+\frac{a}{2})^2+r^2} - (z-\frac{a}{2})} = \frac{\sqrt{\ominus + r^2} - \oplus}{\sqrt{\oplus + r^2} - \ominus}$$

$$\ln \frac{\left(\ominus + \sqrt{r^2+\ominus}\right)\left(\oplus + \sqrt{r^2+\ominus}\right)}{\left(\ominus - \sqrt{r^2+\ominus}\right)\left(\oplus - \sqrt{r^2+\oplus}\right)}$$

$$= \ln \frac{\left(\sqrt{r^2+\ominus} + \ominus\right)\left(\sqrt{r^2+\oplus} + \oplus\right)}{\left(\sqrt{r^2+\ominus} - \ominus\right)^2\left(\sqrt{r^2+\oplus} - \oplus\right)}$$

11.

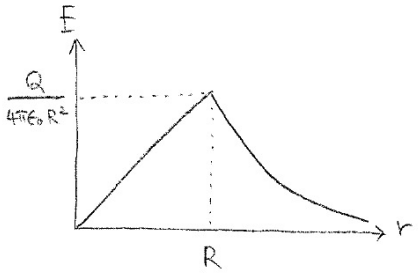
導体球



$0 < r < R \quad E = 0$

$R < r \quad E = \dfrac{Q}{4\pi\varepsilon_0 r^2}$

一様分布

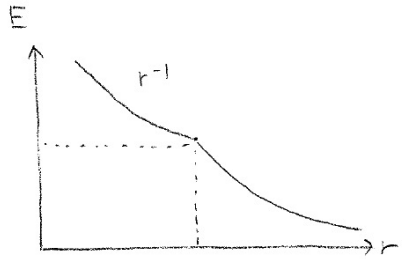


$0 < r < R \quad E = \dfrac{Q}{4\pi\varepsilon_0} \cdot \dfrac{r}{R^3}$

$R < r \quad E = \dfrac{Q}{4\pi\varepsilon_0 r^2}$

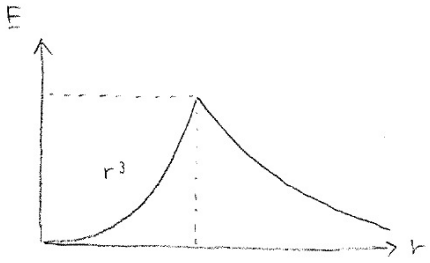
球対称

$n = -2$



$0 < r < R \quad E = \dfrac{Q}{4\pi\varepsilon_0 R^2}\left(\dfrac{r}{R}\right)^{n+1}$

$n = 2$



12. $\phi(r) = \frac{q}{4\pi\epsilon_0 a}\left(1+\frac{a}{r}\right)e^{-\frac{2r}{a}}$

$$E = -\nabla\phi = -\frac{\partial\phi}{\partial r} = \frac{q}{4\pi\epsilon_0 a}\left(\frac{2}{a}+\frac{2}{r}+\frac{a}{r^2}\right)e^{-\frac{2r}{a}}$$

$$\rho = \epsilon_0 \nabla\cdot E = \frac{\epsilon_0}{4\pi r^2}\frac{d}{dr}\left(4\pi r^2 E\right) = -\frac{4\epsilon_0}{a^2}e^{-\frac{2r}{a}}\cdot\frac{q}{4\pi\epsilon_0 a}$$

$$\frac{\partial}{\partial r}\frac{q}{4\pi\epsilon_0 a}\left(\frac{2}{a}r^2+2r+a\right)e^{-\frac{2r}{a}}$$

$$\left\{\frac{4}{a}r+2-\frac{2}{a}\left(\frac{2}{a}r^2+2r+a\right)\right\}$$

$$\frac{4r}{a}+2-\frac{4r^2}{a^2}-\frac{4r}{a}-2$$

$$q = \int \rho\, 4\pi r^2\, dr$$

$$\int \frac{4q}{a^3}r^2 e^{-\frac{2r}{a}}\,dr$$

$$\rho = -\frac{q}{\pi a^3}e^{-\frac{2r}{a}} + q\,\delta(r)$$

$(\nabla^2 \frac{1}{r} = 0 \text{ at } r\neq 0)$

$$\frac{4q}{a^3}\left(\frac{\pi a}{2}\right)^{\frac{1}{2}}$$

$$\frac{q}{4\pi\epsilon_0 a}\frac{\epsilon_0}{r^2}\frac{d}{dr}\left(\frac{2}{a}r^2+2r+a\right)e^{-\frac{2r}{a}}$$

$$= \frac{q}{4\pi a}\frac{1}{r^2}\left\{\left(\frac{4r}{a}+2\right)-\frac{2}{a}\left(\frac{2}{a}r^2+2r+a\right)\right\}e^{-\frac{2r}{a}}$$

$$q =$$

$$= \frac{q}{4\pi a}\frac{1}{r^2}\left(-\frac{4r^2}{a^2}\right)e^{-\frac{2r}{a}}$$

$$= \frac{-q}{\pi a^3}$$

$$\rho = \left(-\frac{q}{\pi a^3}e^{-\frac{2r}{a}}\right.$$

13

Boundary $\begin{cases} \phi(R,\theta) = V & A_0 + B_0/R = V \;,\; A_n + B_n \cdot R^{-2n-1} = 0 \quad (n>0) \\ \phi(r,\theta) \to -E\hat{z} = -E_0 r\cos\theta \end{cases}$

$$\phi = \left(A_0 + \frac{B_0}{r}\right) + E_0\left(-r + \frac{R^3}{r^2}\right)\cos\theta$$

$$\sigma = \epsilon_0 E_r = -\epsilon_0 \frac{\partial\phi}{\partial r} = \epsilon_0 \frac{B_0}{R^2} + 3\epsilon_0 E_0 \cos\theta$$

$$\nabla^2\phi = 0 \;\to\; \phi = \sum_{l=0}^{\infty}\sum_{m=-l}^{l}\left[A_{lm} r^l + B_{lm} r^{-l-1}\right] P_l^{(m)}(\cos\theta)\, e^{im\varphi}$$

14

$$\frac{1}{2}\sum q_i \phi_i = \frac{1}{2}\int \rho\phi\, d\tau = \frac{1}{2}\epsilon_0 \int (\nabla\phi)\cdot(\nabla\phi) - \nabla\cdot(\phi\nabla\phi)\, d\tau$$

$$= \frac{1}{2}\epsilon_0 \int E^2 d\tau + \frac{1}{2}\epsilon_0 \int \phi E \cdot dS$$

$$\int_{surf} \phi E \cdot dS = 0$$

$$\frac{1}{r}\cdot\frac{1}{r^2}\cdot r^2 \to 0$$

15. グリーンの定理

$$\phi_i = \sum_{j \neq i} \frac{1}{4\pi\varepsilon_0} \frac{q_j}{r_{ij}} \qquad \phi_i' = \sum_{j \neq i} \frac{1}{4\pi\varepsilon_0} \frac{q_j'}{r_{ij}}$$

$$\sum_i q_i \phi_i' = \sum_i \sum_{j \neq i} \frac{1}{4\pi\varepsilon_0} \frac{q_i q_j'}{r_{ij}} = \sum_i q_i' \phi_i$$

$$q_1' = -q \frac{d_1}{d_1 + d_2}$$

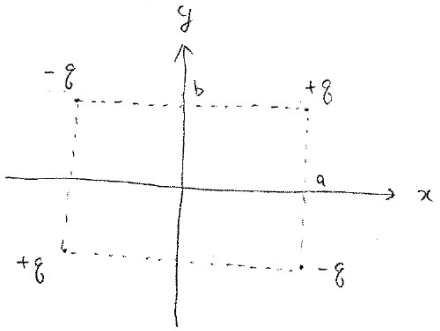
$$q_2' = -q \frac{d_2}{d_1 + d_2}$$



$\sum q_i \phi_i' = \sum q_i' \phi_i \qquad \Delta\phi_i = \frac{1}{\epsilon} q_i$?

$\Leftrightarrow \sum_i (\Delta\phi_i) \phi_i' - \sum_i (\Delta\phi_i') = 0$?

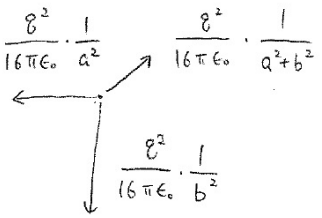
16、電気映像法



$$\Phi = \frac{q}{4\pi\epsilon_0}\left(\frac{1}{R_{--}} + \frac{1}{R_{++}} - \frac{1}{R_{+-}} - \frac{1}{R_{-+}}\right)$$

$$R_{\pm\pm} = \sqrt{(x\pm a)^2 + (y\pm b)^2 + z^2}$$

$\Phi(x=0) = \Phi(y=0) = 0$



$\Rightarrow$

$$\frac{q^2}{16\pi\epsilon_0}\cdot\frac{1}{a^2}\left\{1-\left(1+\frac{b^2}{a^2}\right)^{-3/2}\right\}$$



$$\frac{q^2}{16\pi\epsilon_0}\cdot\frac{1}{b^2}\left\{1-\left(1+\frac{a^2}{b^2}\right)^{-3/2}\right\}$$

17. $V = e\phi\left(\mathbf{r}+\frac{\mathbf{s}}{2}\right) - e\phi\left(\mathbf{r}-\frac{\mathbf{s}}{2}\right)$

$\simeq e\left\{\phi(\mathbf{r}) + \frac{1}{2}\mathbf{s}\cdot\nabla\phi(\mathbf{r})\right\} - e\left\{\phi(\mathbf{r}) - \frac{1}{2}\mathbf{s}\cdot\nabla\phi(\mathbf{r})\right\} = e\mathbf{s}\cdot\nabla\phi(\mathbf{r}) = -\mathbf{p}\cdot\mathbf{E}(\mathbf{r})$

$\therefore \mathbf{F} = \nabla(\mathbf{p}\cdot\mathbf{E})$

$= \underbrace{(\mathbf{p}\cdot\nabla)\mathbf{E}}_{\sim\sim} + (\mathbf{E}\cdot\nabla)\mathbf{p} + \mathbf{p}\times(\nabla\times\mathbf{E}) + \mathbf{E}\times(\nabla\times\mathbf{p})$

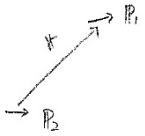
$(\mathbf{E}\cdot\nabla)\mathbf{p} = 0$, $\mathbf{p}\times(\nabla\times\mathbf{E}) = 0$, $\mathbf{E}\times(\nabla\times\mathbf{p}) = 0$

$\frac{e}{4\pi\epsilon_0}\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = \frac{e}{4\pi\epsilon_0}\left(\frac{1}{|\mathbf{r}+\frac{\mathbf{s}}{2}|} - \frac{1}{|\mathbf{r}-\frac{\mathbf{s}}{2}|}\right)$

$$\phi = \frac{e}{4\pi\epsilon_0}\, s\,\frac{\partial}{\partial s}\left(\frac{1}{r}\right) = \mathbf{p}\cdot\frac{1}{4\pi\epsilon_0}\nabla\left(\frac{1}{r}\right) = \frac{\mathbf{p}\cdot\mathbf{r}}{4\pi\epsilon_0 r^3}$$

$$\mathbf{E} = -\nabla_r\phi = -\frac{1}{4\pi\epsilon_0}\left[\mathbf{r}\cdot\mathbf{p}\,\nabla_r\left(\frac{1}{r^3}\right) + \frac{1}{r^3}\nabla_r(\mathbf{r}\cdot\mathbf{p})\right]$$

$$= \frac{1}{4\pi\epsilon_0}\left[\frac{3(\mathbf{r}\cdot\mathbf{p})\mathbf{r} - r^2\mathbf{p}}{r^5}\right] - \frac{1}{4\pi\epsilon_0}\cdot\frac{4\pi}{3}\mathbf{p}\cdot\delta(\mathbf{r})$$



相互作用エネルギー

$$U = -\mathbf{P}_1\cdot\frac{1}{4\pi\epsilon_0}\cdot\frac{3(\mathbf{r}\cdot\mathbf{P}_2)\mathbf{r} - r^2\mathbf{P}_2}{r^5} = \frac{1}{4\pi\epsilon_0}\left[-\frac{3}{r^5}(\mathbf{r}\cdot\mathbf{P}_1)(\mathbf{r}\cdot\mathbf{P}_2) + \frac{1}{r^3}(\mathbf{P}_1\cdot\mathbf{P}_2)\right]$$

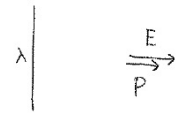
力 $\mathbf{F} = -\nabla_r U = \frac{1}{4\pi\epsilon_0}\left[-3(\mathbf{r}\cdot\mathbf{P}_1)(\mathbf{r}\cdot\mathbf{P}_2)\nabla_r\left(\frac{1}{r^5}\right) - \frac{3}{r^5}\nabla_r\left\{(\mathbf{r}\cdot\mathbf{P}_1)(\mathbf{r}\cdot\mathbf{P}_2)\right\}\right]$

$$= \frac{1}{4\pi\epsilon_0}\left[\frac{15(\mathbf{r}\cdot\mathbf{P}_1)(\mathbf{r}\cdot\mathbf{P}_2)\mathbf{r}}{r^7} - \frac{3\left\{(\mathbf{r}\cdot\mathbf{P}_2)\mathbf{P}_1 + (\mathbf{r}\cdot\mathbf{P}_1)\mathbf{P}_2\right\}}{r^5}\right]$$

18.

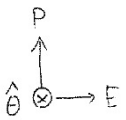
$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

(1) parallel　　トルク　$\mathbb{N} = \mathbb{P} \times \mathbb{E} = 0$



力　$\mathbb{F} = \nabla(\mathbb{P} \cdot \mathbb{E}) = -\frac{\lambda P}{2\pi\epsilon_0 r^3}$

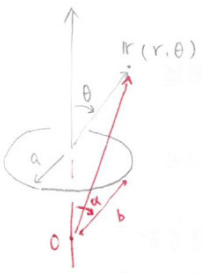
(2) perpendicular　　トルク　$\mathbb{N} = \mathbb{P} \times \mathbb{E} = \frac{\lambda P}{2\pi\epsilon_0 r}\hat{\theta}$





力　$\mathbb{F} = \nabla(\mathbb{P} \cdot \mathbb{E}) = 0$

19



$\nabla^2\phi = -\frac{\lambda}{\epsilon}\delta(\theta)\delta(r-a) \rightarrow \nabla^2\phi = 0$ の境界値問題

$$\phi = R(r)\Theta(\theta) = \frac{q}{4\pi\epsilon_0}\sum_{l=0}^{\infty}\left[A_l r^l + B_l r^{-(l+1)}\right]P_l(\cos\theta)$$

$$\frac{\Theta(\theta)}{r^2}\frac{d}{dr}\left(r^2R'(r)\right) + \frac{R(r)}{r^2\Lambda\Theta}\frac{d}{d\theta}\left(\Lambda\Theta\,\Theta'(\theta)\right) + \frac{\lambda}{\epsilon}\delta(\theta)\delta(r-a) = 0$$

$$\frac{1}{r^2R}\frac{d}{dr}\left(r^2R'\right) - \frac{1}{r^2\Theta}\frac{d}{d\mu}\left((1-\mu^2)\frac{d\Theta}{d\mu}\right) + \frac{\lambda}{\epsilon}\cdot\frac{\delta(\theta)}{\Theta(\theta)}\cdot\frac{\delta(r-a)}{R(r)} = 0$$

$$\Phi(r,\theta=0) = \frac{q}{4\pi\epsilon_0}\cdot\frac{1}{\sqrt{r^2+b^2-2rb\cos\alpha}}$$

$$(1-2z\cos\theta+z^2)^{-1/2} = \sum_{l=0}^{\infty}P_l(\cos\theta)\,z^l$$

$$= \begin{cases} \frac{q}{4\pi\epsilon_0}\cdot\frac{1}{b}\sum\left(\frac{r}{b}\right)^l P_l(\cos\alpha) & r<b \\ \frac{q}{4\pi\epsilon_0}\cdot\frac{1}{r}\sum\left(\frac{b}{r}\right)^l P_l(\cos\alpha) & r>b \end{cases}$$

$$\phi(r,\theta) = \begin{cases} \frac{\varepsilon}{4\pi\epsilon}\cdot\frac{1}{b}\sum\left(\frac{r}{b}\right)^l P_l(\cos\alpha)P_l(\cos\theta) & r<b \\ \frac{\varepsilon}{4\pi\epsilon}\cdot\frac{1}{r}\sum\left(\frac{b}{r}\right)^l P_l(\cos\alpha)P_l(\cos\theta) & r>b \end{cases}$$

$$A_l = \frac{1}{b^{l+1}}P_l(\cos\alpha)$$

20 Boundary $\begin{cases} \epsilon \frac{\partial \phi}{\partial r} = \epsilon \frac{\partial \phi_0}{\partial r} \quad , \quad \frac{\partial \phi}{\partial \theta} = \frac{\partial \phi_0}{\partial \theta} \\ \phi(0,\theta) = 0 \end{cases}$

$$\phi(r,\theta) = A_1 r\cos\theta = -\frac{\epsilon_0}{\epsilon + 2\epsilon_0} E_0 r\cos\theta$$

$$\phi_0(r,\theta) = \left(A_1' r + \frac{B_1}{r^2}\right)\cos\theta = -E_0 r\cos\theta + \frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \cdot \frac{a^3}{r^2} E_0 \cos\theta$$

21 $\phi_{in} = Ar\mu$

$\phi = \frac{B}{r} + (Cr + \frac{D}{r^2})\mu$

$\phi_{ext} = \frac{F}{r} + (-E_0 r + \frac{G}{r^2})\mu$ ☺ $\phi_{ext} \sim -E_0 r\mu$

$\frac{\partial}{\partial\mu}|_{R_1}$ $AR_1 = CR_1 + \frac{D}{R_1^2}$

$\frac{\partial}{\partial\mu}|_{R_2}$ $CR_2 + \frac{D}{R_2^2} = -E_0 R_2 + \frac{G}{R_2^2}$

$\epsilon\frac{\partial}{\partial r}|_{R_1}$ $\epsilon_0 A\mu = -\frac{\epsilon B}{R_1^2} + \epsilon\left(C - \frac{2D}{R_1^3}\right)\mu$

$\epsilon\frac{\partial}{\partial r}|_{R_2}$ $-\frac{\epsilon B}{R_2^2} + \epsilon\left(C - \frac{2D}{R_2^3}\right)\mu = -\frac{\epsilon_0 F}{R_2^2} + \epsilon_0\left(-E_0 - \frac{2G}{R_2^3}\right)\mu$

$$\begin{cases} A = C + \frac{D}{R_1^3}, \quad C + \frac{D}{R_2^3} = -E_0 + \frac{G}{R_2^3} \\ B = 0, \quad \epsilon_0 A = \epsilon\left(C - \frac{2D}{R_1^3}\right), \quad F = 0, \quad \epsilon\left(C - \frac{2D}{R_2^3}\right) = \epsilon_0\left(-E_0 - \frac{2G}{R_2^3}\right) \end{cases}$$

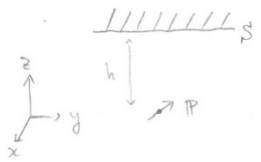
$$D = \frac{3\epsilon_0(\epsilon_0 - \epsilon)E_0}{\frac{(2\epsilon_0 - \epsilon)(\epsilon_0 + 2\epsilon)}{R_1^3} - \frac{2(\epsilon_0 - \epsilon)(\epsilon_0 + \epsilon)}{R_2^3}}$$

$$C = \frac{-3\epsilon_0(\epsilon_0 + 2\epsilon)E_0}{(2\epsilon_0 - \epsilon)(\epsilon_0 + 2\epsilon) - 2(\epsilon_0 - \epsilon)(\epsilon_0 + \epsilon)\frac{R_1^3}{R_2^3}}$$

$$A = \frac{-3\epsilon_0\epsilon E_0}{(2\epsilon_0 - \epsilon)(\epsilon_0 + 2\epsilon) - 2(\epsilon_0 - \epsilon)(\epsilon_0 + \epsilon)\frac{R_1^3}{R_2^3}}$$

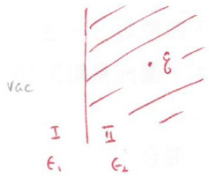
$$G = -\frac{(\epsilon_0 + \epsilon)(\epsilon_0 + 2\epsilon)R_2^3 + (\epsilon_0 - \epsilon)(5\epsilon_0 + 2\epsilon)R_1^3}{(2\epsilon_0 - \epsilon)(\epsilon_0 + 2\epsilon)R_2^3 - 2(\epsilon_0 - \epsilon)(\epsilon_0 + \epsilon)R_1^3} \cdot R_2^3 E_0$$

22.



Pのつくるポテンシャル $\phi = \frac{P \cdot r}{4\pi\epsilon_0 r^3}$

Boundary
$$\begin{cases} \epsilon_0 \frac{\partial \phi_0}{\partial z} = \epsilon \frac{\partial \phi}{\partial z} \quad (z=h) \\ \frac{\partial \phi_0}{\partial x} = \frac{\partial \phi}{\partial x}, \quad \frac{\partial \phi_0}{\partial y} = \frac{\partial \phi}{\partial y} \end{cases}$$

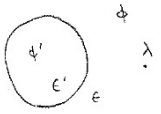


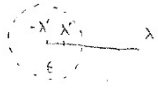
$$\phi_2(P) = \frac{1}{4\pi\epsilon_0}\left(\frac{\varepsilon}{\epsilon_2 R} - \frac{\varepsilon'}{R'}\right)$$

$$\phi_1(P) = \frac{1}{4\pi\epsilon_0}\cdot\frac{\varepsilon''}{R}$$



23.



φ ⌋ 

$$\phi = -\frac{1}{2\pi\epsilon}\left(\lambda \ln R + \lambda' \ln R' - \lambda' \ln r\right)$$

φ' ⌋ 

$$\phi' = -\frac{1}{2\pi\epsilon'}\lambda'' \ln R$$

$$R^2 = b^2 + r^2 - 2br\cos\theta$$

$$R'^2 = \left(\frac{a^2}{b}\right)^2 + r^2 - 2\frac{a^2}{b} r\cos\theta$$

$$\phi = \phi' \rightarrow \left(b^2 + a^2 - 2ba\cos\theta\right)^{\frac{1}{2}\left(\frac{\lambda}{\epsilon} - \frac{\lambda''}{\epsilon'}\right)} \cdot \left(\frac{a^4}{b^2} + a^2 - 2\frac{a^3}{b}\cos\theta\right)^{\frac{\lambda'}{2\epsilon}} = a^{\frac{\lambda'}{\epsilon}}$$

$$\left(\frac{a^2}{b^2} + 1 - 2\frac{a}{b}\cos\theta\right)^{\frac{1}{2}\left(\frac{\lambda+\lambda'}{\epsilon} - \frac{\lambda''}{\epsilon'}\right)} = b^{\left(\frac{\lambda''}{\epsilon'} - \frac{\lambda}{\epsilon}\right)}$$

$$\frac{\lambda+\lambda'}{\epsilon} - \frac{\lambda''}{\epsilon'} = 0 \quad \cdots (1)$$

$$\epsilon\frac{\partial\phi}{\partial r} = \epsilon'\frac{\partial\phi'}{\partial r} \rightarrow (\lambda - \lambda'')\frac{a - b\cos\theta}{a^2 + b^2 - 2ab\cos\theta} + \lambda'\frac{a - \frac{a^2}{b}\cos\theta}{a^2 + \frac{a^4}{b^2} - 2\frac{a^3}{b}\cos\theta} = \frac{\lambda'}{a}$$

$$\frac{a}{b}\cdot\frac{\lambda - \lambda''}{\lambda'}\left(\frac{a}{b} - \cos\theta\right) + \left(1 - \frac{a}{b}\cos\theta\right) = \left(\frac{a^2}{b^2} + 1 - 2\frac{a}{b}\cos\theta\right)$$

$$\frac{a^2}{b^2}\left(\frac{\lambda - \lambda''}{\lambda'} - 1\right) - \frac{a}{b}\left(\frac{\lambda - \lambda''}{\lambda'} - 1\right)\cos\theta = 0$$

$$\lambda - \lambda'' = \lambda' \quad \cdots (2)$$

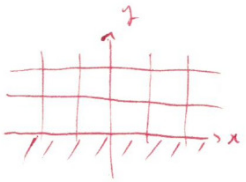
$$\therefore \begin{cases} \lambda' = \dfrac{\epsilon - \epsilon'}{\epsilon + \epsilon'}\lambda \\ \lambda'' = \dfrac{2\epsilon'}{\epsilon + \epsilon'}\lambda \end{cases}$$

単位長さあたりに働く力

$$f = \frac{1}{2\pi\epsilon}\left(\frac{\lambda'}{b - \frac{a^2}{b}} - \frac{\lambda'}{b}\right) = \frac{1}{2\pi\epsilon}\cdot\frac{\lambda'}{b}\cdot\frac{1}{\frac{b^2}{a^2} - 1}$$

$$\frac{b - b + \left(\frac{a^2}{b}\right)}{b^2 - a^2}$$

24



analytic



ω

$\omega = z^{\frac{\alpha}{\pi}}$

($z \neq 0$ 之等角)

$x = a \rightarrow \omega = (a+iy)^{\frac{\alpha}{\pi}}$ 電場力線 ①

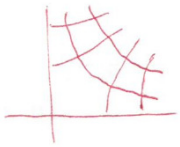
$y = b \rightarrow \omega = (x+ib)^{\frac{\alpha}{\pi}}$ 等電位面 ②

$re^{i\theta}$

① $\rightarrow a+iy = r^{\frac{\pi}{\alpha}} e^{\frac{i\theta\pi}{\alpha}}$

$a = r^{\frac{\pi}{\alpha}} \cos\frac{\theta\pi}{\alpha}$

$b = r^{\frac{\pi}{\alpha}} \sin\frac{\theta\pi}{\alpha}$



$a = r^2(x^2-y^2)$

$b = 2r^2xy$

$\boxed{\Delta\phi = 0}$; $f(z)$ 正則

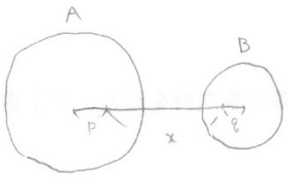
$= u+iw$

$(\Delta u = \Delta w = 0)$

$f(z) = \phi + i\psi$

$f(z) \rightarrow g(z)$

25



$$\begin{cases} x+p+q=r \\ \dfrac{a^2}{p+x}=p \\ \dfrac{b^2}{q+x}=q \end{cases}$$

$$p=\frac{-x+\sqrt{x^2+4a^2}}{2}$$

$$\sqrt{x^2+4a^2}+\sqrt{x^2+4b^2}=2r$$

$a^2=p(r-q)$

$b^2=q(r-p)$

$a^2+b^2=(p+q)r-2pq$

$a^2-b^2=(p-q)r$

$\frac{a^2}{p}=p+x$

$\frac{b^2}{q}=q+x$

$$U=\frac{\epsilon_0}{2}\int|E|^2d\tau=\frac{1}{2}\int(\rho_A+\rho_B)(\phi_A+\phi_B)d\tau=\frac{1}{2}\int_{V_A}\rho_A\phi_A d\tau+\frac{1}{2}\int_{V_B}\rho_B\phi_B d\tau+\int_{V_B}\rho_B\phi_A d\tau$$

↑

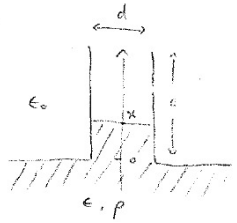
(Green 함수 적분정리 $\int_{V_A}\rho_A\phi_B d\tau=\int_{V_B}\rho_B\phi_A d\tau$)

$$\int_{V_B}\rho_B\phi_A d\tau=\frac{\rho_B Q_A}{3\epsilon_0 r}b^3$$

$$\int_{V_A}\rho_A\phi_A dV=\frac{3Q_A^2}{16\pi\epsilon_0 a}$$

$$U=\frac{1}{4\pi\epsilon_0}\left(\frac{3}{5}\left(\frac{Q_A^2}{a}+\frac{Q_B^2}{b}\right)-\frac{Q_AQ_B}{r}\right)$$

26.



$$C = \epsilon_0 \frac{a(a-x)}{d} + \epsilon \frac{ax}{d} = \frac{a}{d}\left(\epsilon_0 a + (\epsilon - \epsilon_0)x\right)$$

$$U = \frac{1}{2} C V^2$$

$$F_x dx + \frac{V^2}{2} dC = V^2 dC$$



$$F_x dx + dU = V\,dQ \rightarrow F_x = \frac{V^2}{2}\frac{dC}{dx} = \frac{(\epsilon-\epsilon_0)aV^2}{2d}$$

$$\int_0^x \rho g x' \underbrace{dV}_{a\,d\,dx'}$$

$$\rho (a\,dx)\ddot{x} = \frac{(\epsilon-\epsilon_0)aV^2}{2d} - \rho(a\,dx)g$$

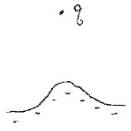
$\rho g a d \frac{x^2}{2}$

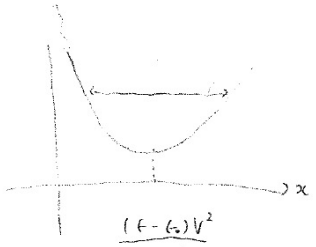
M

$Mg\frac{x}{2}$

$$\ddot{x} = \frac{(\epsilon-\epsilon_0)V^2}{2\rho d^2}\cdot\frac{1}{x} - g$$

$$\frac{1}{2}\dot{x}^2 - \frac{(\epsilon-\epsilon_0)V^2}{2\rho d^2}\ln x + gx = \text{const}$$

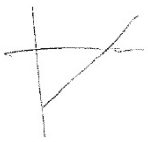




$$d\left(\frac{1}{2}M\dot{x}^2\right) + d(Mgx) = F_x\,dx$$

$$= \frac{1}{2}V^2 \frac{a}{d}(\epsilon-\epsilon_0)\,dx$$

$d($



$$\frac{1}{2}\dot{x}^2 + gx - \frac{V^2}{2}\frac{\frac{a}{d}\left((\epsilon-\epsilon_0)x + \epsilon_0 a\right)}{x\,a\,d} = \text{const}$$

$$\frac{1}{2}x\dot{x}^2 + gx^2 - \frac{V^2}{2d^2\rho}\left((\epsilon-\epsilon_0)x + \epsilon_0 a\right) = \text{const}$$

~~$\frac{1}{2}\dot{x}^2 + x\dot{x}\ddot{x}$~~

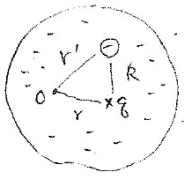
$$\frac{1}{2}\dot{x}^3 + x\dot{x}\ddot{x} + 2gx\dot{x} - \frac{(\epsilon-\epsilon_0)V^2}{2\rho d^2}\dot{x} = 0$$

$$\frac{1}{2}\dot{x}^2 + x\ddot{x} + 2gx - \frac{(\epsilon-\epsilon_0)V^2}{2\rho d^2} = 0$$

$$\frac{1}{2}\dot{x}^2 + gx - \frac{(\epsilon-\epsilon_0)V^2}{2\rho a^2} = \frac{\text{const}}{x}$$

$$\frac{1}{2}\dot{x}^2 + \frac{1}{2}gx - \frac{(\epsilon-\epsilon_0)V^2}{2\rho d^2} = 0$$

27



$$\phi(r) = \frac{1}{4\pi\epsilon_0}\int \frac{\rho}{R} r'^2 \sin\theta \, d\theta dr' \cdot 2\pi$$

$$= \frac{\rho}{6\epsilon_0}(3a^2 - r^2)$$

$$R^2 = (r^2 + r'^2 - 2rr'\cos\theta)$$

$$E_-(r) = \frac{\rho}{3\epsilon_0} r$$

$$|E| = |E_-| \rightarrow qr = 4\pi\epsilon_0 a^3 E$$

$$\vec{p} = nqr$$

$$\epsilon = \epsilon_0 + \frac{nqr}{E} = \epsilon_0(1 + 4\pi a^3 n)$$

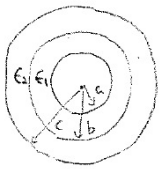
$$E_- = \frac{\rho}{3\epsilon_0} r \qquad |E_-| = |E| \qquad \frac{1}{4\pi\epsilon_0}\frac{q}{a^3} = \frac{E}{r}$$

$$\vec{p} = \alpha \vec{E} = nqr$$

$$E_- = \frac{\rho}{3\epsilon_0} r = \frac{q}{4\pi\epsilon_0} \cdot \frac{r}{a^3}$$

$$\langle E \rangle =$$

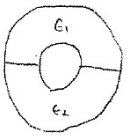
28.



$$E = \frac{Q}{4\pi\epsilon r^2}$$

$$V = \frac{Q}{4\pi\epsilon_1}\int_a^b \frac{dr}{r^2} + \frac{Q}{4\pi\epsilon_2}\int_b^c \frac{dr}{r^2} = \frac{Q}{4\pi\epsilon_1}\left(\frac{1}{a}-\frac{1}{b}\right) + \frac{Q}{4\pi\epsilon_2}\left(\frac{1}{b}-\frac{1}{c}\right)$$

$$\therefore C = \frac{Q}{V} = \left\{ \frac{\frac{1}{a}-\frac{1}{b}}{4\pi\epsilon_1} + \frac{\frac{1}{b}-\frac{1}{c}}{4\pi\epsilon_2} \right\}^{-1}$$



$$2\pi r^2(\epsilon_1+\epsilon_2)E = Q$$



採用法と仮定 → 解一つ → ちゃうえ解式をみたす唯一つの解ゆえ O.K.

[30] $M = \frac{1}{2}\int \vec{r} \times \vec{j}\, d\tau = \frac{1}{2} r e v = \frac{e}{2m} L \qquad (L = mrv)$

$m\ddot{x} = -\frac{\partial U}{\partial x} + e\dot{x} \times B$ Zeemann 效果

$\frac{d}{dt}\left(\frac{1}{2} m |\dot{x}|^2 + U\right) = 0$

$\frac{d}{dt}\left(L + \frac{1}{2} e B |x|^2\right) = e(x \cdot B)\cdot \dot{x}$ $\quad L = x \times p$

$\dot{L} = e x \times (\dot{x} \times B)$



$E \cdot 2\pi r = -\pi r^2 \frac{dB}{dt}$

$E = -\frac{1}{2} r \frac{dB}{dt}$

$m\frac{dv}{dt} = -eE = \frac{1}{2} er \frac{dB}{dt}$

$v' - v = \frac{er}{2m}(B - 0)$

$\Delta M = \frac{1}{2} r e \Delta v = \frac{r^2 e^2}{4m} B$

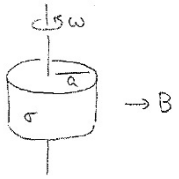


$\vec{F} = \vec{M} \times \vec{B}$

$\frac{d\vec{L}}{dt} = \vec{M} \times \vec{B}$

[34]

$J = \sigma E$



$dI = J r dr d\theta$

$dN = r\cos\theta \, B \, dI$

$N = -\sigma B^2 \omega \int r^3 \cos^2\theta \, dr d\theta$

$E = -vB\cos\theta$

$= -r\omega B\cos\theta$

$= -\frac{\pi}{4}\sigma B^2 \omega a^4$

$J = \sigma E = -\sigma\omega r B\cos\theta$

$I\frac{d\omega}{dt} = N$ $\quad \omega = \omega_0 e^{-kt}$

$k = \frac{\pi}{4}\sigma B^2 a^4 I^{-1}$

$I \propto a^4$

$\vec{r} \times \left[(\vec{v} \times \vec{B}) \times \vec{B} \right]$ (-θ)

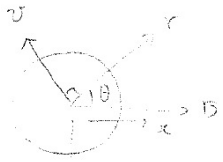


$\vec{J} \times \vec{B} = \sigma r\omega B\cos\theta \times \vec{B}$ (-r̂)

$= -\sigma r\omega B^2 \cos\theta \, \hat{\theta}$

$\vec{F} = q\vec{v} \times \vec{B}$

$= -\vec{v} \times \vec{B}$



$-\vec{v}$

$\vec{r} \, r dr d\theta \times \vec{B}$

$\sigma r dr d\theta \, d\vec{E} \times \vec{B}$

$I = \int \rho r^2 dV$ $\quad (r dr d\theta)$

$\vec{r} \times (\vec{J} \, r dr d\theta \times \vec{B})$

$= \vec{r} \times (\vec{J} \times \vec{B}) \, r dr d\theta$

$= \vec{r} \times (\vec{E} \times \vec{B}) \, \sigma r \, dr d\theta$

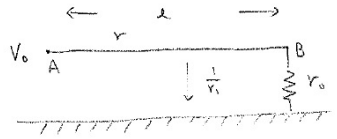
$\int \rho r^3 dr d\theta$

$\rho \frac{a^4}{4} \cdot 2\pi$

$\frac{\pi}{2}\rho a^4 \omega' = -\frac{\pi}{4}\sigma B^2 \omega a^4$

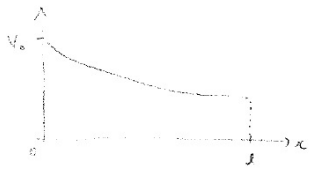
$\omega' = -\frac{\sigma}{\rho} B^2 \omega$

[36]



$\dot{d} = \sigma E$

$I_{leak} = \frac{1}{r_1} \cdot V(x)$ $\qquad E(x) = \frac{dV}{dx}$

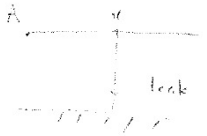


$$\left(V_0 = \int_0^l E\,dx + r_0 I(x) \right)$$

$$V_0 = r_0 I_0 + \int_0^l (r I_0 + I_{leak})\,dx$$

$$\left(V_0 = \int_0^l r\, I(x)\,dx + r_0 I(0) \right.$$

$$\left. V_0 = \int_0^x r\, I(x')\,dx' + \left(-r_1 \frac{dI}{dx}\right) \right)$$



$$r\,I(x) - r_1 \frac{d^2 I}{dx^2} = 0$$

$$I = A e^{\sqrt{\frac{r}{r_1}}x} + B e^{-\sqrt{\frac{r}{r_1}}x}$$

$$\frac{dI}{dx} = \sqrt{\frac{r}{r_1}} \left(A e^{\sqrt{\frac{r}{r_1}}x} - B e^{-\sqrt{\frac{r}{r_1}}x} \right) \quad \text{(x=0)} \rightarrow \quad -\frac{V_0}{r_1} = \sqrt{\frac{r}{r_1}}\,(A - B)$$

$$V_0 = \int_0^x r\, I(x')\,dx' =$$

[38]

$$m\ddot{\vec{x}} = q\,\dot{\vec{x}} \times \vec{B}$$

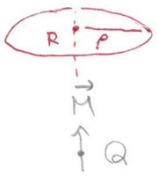
$$\begin{cases} \ddot{x} = \frac{qB}{m}\dot{y} \\ \ddot{y} = -\frac{qB}{m}\dot{x} \\ \ddot{z} = 0 \quad \rightarrow \quad z = \alpha t + \beta \end{cases}$$

$$w = \dot{x} + i\dot{y} \quad \rightarrow \quad \ddot{w} = -\left(\frac{qB}{m}\right)^2 w$$

$$\dot{x} + i\dot{y} = Ae^{i\omega t} + Be^{-i\omega t} \qquad \omega = \frac{qB}{m}$$

$$x + iy = Ae^{i\omega t} + Be^{-i\omega t} + C$$

[37]



$$\vec{N} = \vec{M} \times \vec{B}$$

$$U(r) = -\vec{M} \cdot \vec{B}$$

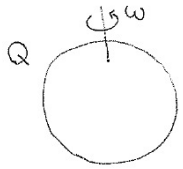
$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{r}$$

$$\vec{B} = \nabla \frac{1}{4\pi} \frac{\vec{\mu} \cdot \vec{r}}{r^3} = \frac{1}{4\pi} \left(\frac{\vec{\mu}}{r^3} - \frac{3(\vec{\mu} \cdot \vec{r})\,\vec{r}}{r^5} \right)$$

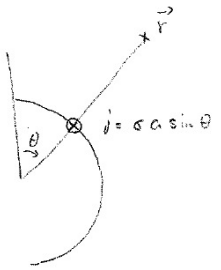
$$m\ddot{\vec{r}} = q(\vec{E} + \dot{\vec{r}} \times \vec{B})$$

$$m\ddot{\vec{\rho}} = \frac{qQ}{4\pi\epsilon_0 r^3}(\vec{R} + \vec{\rho}) + \frac{q}{4\pi}\left(\frac{\dot{\vec{\rho}} \times \vec{\mu}}{r^3} - \frac{3(\vec{\mu} \cdot \vec{r})}{r^5} \right.$$

[39]



$$A(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{j}(\vec{r}')}{R} \, dV'$$



$4\pi a^2 \sigma = Q$

$$M = \frac{4}{5}\pi \sigma \omega a^4 = \frac{1}{3} Q \omega a^2 \qquad a \gg a$$

$$B = \frac{M}{4\pi\varepsilon_0 c^2} \begin{pmatrix} 3xz \\ 3yz \\ R^2 - 3z^2 \end{pmatrix} \frac{1}{R^5}$$

$$M = \frac{1}{2} \int \vec{r}' \times \vec{j}(\vec{r}') \, dV' \qquad a^3 \sin\theta \, d\theta \, d\varphi$$

$= \frac{1}{2}$



$$\Delta^3\theta = \Delta\theta (1 - \cos^2\theta)$$

$$1 - \int_0^\pi \sin\theta \, d\cos\theta$$

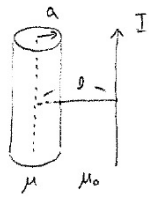
$$\left(\cos\theta - \frac{1}{3}\cos^3\theta \right)_0^\pi$$

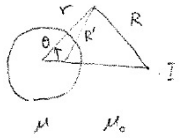
$$(-1 - 1) - \frac{1}{3}(-1 - 1)$$

$$-2 + \frac{2}{3} = -2\left(1 - \frac{1}{3}\right)$$

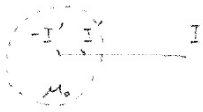


[40]





A₀ :



$$A_0 = -\frac{\mu_0}{2\pi}\left(I \ln R + I' \ln R' - I' \ln r\right)$$

$$\boxed{A = -\frac{\mu_0 I}{2\pi}\ln r}$$

A :



$$A = -\frac{\mu}{2\pi} I'' \ln R$$

$$R^2 = l^2 + r^2 - 2lr\cos\theta$$

$$R'^2 = \left(\frac{a^2}{l}\right)^2 + r^2 - 2\frac{a^2}{l} r\cos\theta \qquad R'^2 = \left(\frac{a}{l}\right)^2 R^2$$

Boundary $(\forall \eta)$ 連続条件 Bの接線成分 Hの接線成分

$$\frac{1}{\mu}\frac{\partial}{\partial\mu}\Big|_a \qquad I\frac{al}{R^2} + I'\frac{a^3/l}{R'^2} = I''\frac{al}{R^2} \longrightarrow \underbrace{\left(I + I' - I''\right)}\frac{al}{a^2+l^2-2al\eta} = 0$$

$$\frac{\partial}{\partial r}\Big|_a \qquad \mu_0 I\frac{a-l\eta}{R^2} + \mu_0 I'\frac{a-\frac{a^2}{l}\eta}{R'^2} - \frac{\mu_0 I'}{a} = \mu I''\frac{a-l\eta}{R^2}$$

$$\longrightarrow \left(\mu_0 I - \mu I''\right)(a^2 - al\eta) + \mu_0 I'\left(l^2 - al\eta\right) = \mu_0 I' R^2$$

$$\longrightarrow \left(\mu_0 I - \mu I''\right)(a^2 - al\eta) = \mu_0 I'(a^2 - al\eta)$$

$$\begin{cases} I' = \dfrac{\mu_0 - \mu}{\mu_0 + \mu} I \\ I'' = \dfrac{2\mu_0}{\mu_0 + \mu} I \end{cases}$$

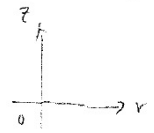
$$f = IB = \frac{\mu_0 I}{2\pi}\left(-\frac{I'}{l} + \frac{I'}{l - \frac{a^2}{l}}\right) = \frac{\mu_0 - \mu}{\mu_0 + \mu}\cdot\frac{\mu_0 I^2}{2\pi}\cdot\frac{a^2/l}{l^2 - a^2} \quad (\text{引力})$$

[41]

$$\vec{B} = \nabla \times \vec{A} = -\hat{r}\frac{\partial A(r,z)}{\partial z} + \hat{z}\frac{1}{r}\frac{\partial}{\partial r}(rA(r,z)) = -\hat{r}\frac{\partial A}{\partial z} + \hat{z}\left(\frac{A}{r} + \frac{\partial A}{\partial r}\right)$$

$$\frac{dz}{dr} = \frac{B_z}{B_r}$$

$d\vec{r} \parallel \vec{B}$

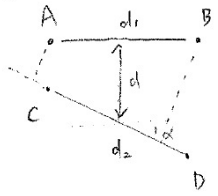


$$\begin{pmatrix} dr \\ d\phi \\ dz \end{pmatrix} \times \begin{pmatrix} B_r \\ 0 \\ B_z \end{pmatrix} = \begin{pmatrix} -B_z\,d\phi \\ B_z\,dr - B_r\,dz \\ -B_r\,d\phi \end{pmatrix}$$

$$B_z\,dr = B_r\,dz$$

$$\left(\frac{A}{r} + \frac{\partial A}{\partial r}\right)dr = -\frac{\partial A}{\partial z}\,dz$$

[42]



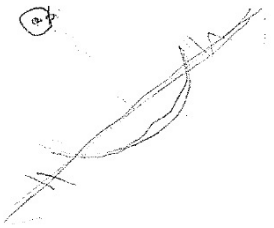
$$L_{12} = \frac{\mu_0}{4\pi}\int_{C_1}\int_{C_2}\frac{d\vec{r}_2 \cdot d\vec{r}_1}{|\vec{r}_1 - \vec{r}_2|}$$

Aから流れる電流Iが CDを切る積分

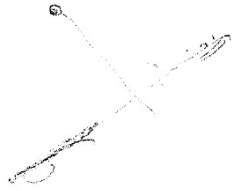
$$\int_{AC}^{AD}\frac{I}{2\pi r}dr$$ Bからの割合は逆符号で $\tan\alpha_0 = \frac{d_1}{2d}$

① $\alpha < \alpha_0$ $\int_{BD}^{BC}\frac{I}{2\pi r}dr$ $\quad \Phi = \frac{\mu I}{2\pi}\ln\frac{AD\cdot BD}{AC\cdot BC}$

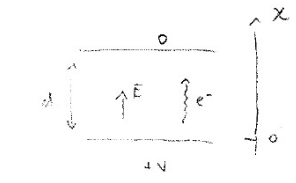
② $\alpha > \alpha_0$ $\int_{BC}^{BD}\frac{I}{2\pi r}dr$ $\quad \Phi = \frac{\mu I}{2\pi}\ln\frac{AB\ BC}{AC\ BD}$







(43)



$n(x)$, $v(x)$, $E(x)$

$$\begin{cases} \dfrac{\partial v}{\partial t} + v\dfrac{\partial v}{\partial x} = -\dfrac{e}{m}E \qquad (\text{Euler}) \\ \dfrac{\partial E}{\partial x} = -\dfrac{e}{\varepsilon_0}n(x) \qquad \leftarrow \quad \nabla\cdot\mathbf{E} = \dfrac{\rho}{\varepsilon_0} \\ e\,n\,v = I \ (\text{const}) \end{cases}$$

c.t. $m\dot{v} = -eE$ (Lagrange)

$$\begin{cases} v\dfrac{\partial v}{\partial x} = -\dfrac{e}{m}E \qquad \rightarrow \quad \dfrac{\partial}{\partial x}\left(\dfrac{1}{2}v^2\right) = -\dfrac{e}{m}E \\ \dfrac{\partial E}{\partial x} = -\dfrac{I}{\varepsilon_0}\cdot\dfrac{1}{v} \end{cases}$$

$$\frac{\partial^2}{\partial x^2}\left(\frac{1}{2}v^2\right) = \frac{2eI}{m\varepsilon_0}\left(\frac{1}{2}v^2\right)^{-\frac{1}{2}}$$

[45] $\vec{B} = \frac{\mu_0 g}{4\pi r^2} e_r$

$\left(\vec{F} = q\,\vec{v} \times \vec{B} = \frac{\mu_0 g q}{4\pi r^2} \left(-r\dot{\theta}\, e_\varphi + r\dot{\varphi}\sin\theta\, e_\theta \right) \right)$

$m \frac{d\vec{v}}{dt} = \frac{\mu_0 g q}{4\pi r^3} \vec{v} \times \vec{r} =$

$m \frac{d\vec{v}}{dt} \cdot \vec{v} = 0 \rightarrow \frac{m}{2} |\vec{v}|^2 = \text{const with } t$

$\vec{r} \times m \frac{d\vec{v}}{dt} = \frac{\mu_0 g q}{4\pi r^3} \left(|\vec{r}|^2 \vec{v} - (\vec{r} \cdot \vec{v}) \vec{r} \right)$

$\frac{d\vec{L}}{dt}$

[47]

$\Box \vec{A} = \nabla^2 \vec{A} - \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2}$ $\qquad \vec{A} = \vec{a} \cos(\vec{k} \cdot \vec{x} - \omega t)$

$= \left(-|\vec{k}|^2 + \frac{\omega^2}{c^2}\right) \vec{A} = 0$

$\therefore |\vec{k}| = \frac{\omega}{c}$

$\mathrm{div}\, \vec{A} + \frac{1}{c^2} \frac{\partial \phi}{\partial t} = 0 \quad \rightarrow \quad \mathrm{div}\, \vec{A} = -\vec{a} \cdot \vec{k} \sin(\vec{k} \cdot \vec{x} - \omega t) = 0$

$\therefore \vec{a} \cdot \vec{k} = 0$

$E = -\frac{\partial A}{\partial t} = \omega \vec{A}$

$B = \nabla \times A = \vec{k} \times \vec{A}$

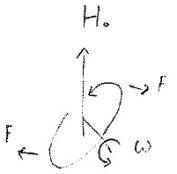
$\vec{Y} = \vec{E} \times \vec{H} = \omega \vec{A} \times \left(\frac{1}{\mu_0} \vec{k} \times \vec{A}\right) = \frac{\omega}{\mu_0} |\vec{A}|^2 \vec{k} - \frac{\omega}{\mu_0} (\vec{A} \cdot \vec{k}) \vec{A}$ ($\vec{A} \cdot \vec{k} = 0$)

$= \frac{\omega}{\mu_0} |\vec{A}|^2 \vec{k}$ // 運動量密度

~~$\vec{p} = \varepsilon_0 \vec{E} \times \vec{B} = \varepsilon_0 \omega |\vec{A}|^2 \vec{k}$~~

エネルギー $W = \frac{\varepsilon_0}{2} \vec{E}^2 + \frac{1}{2\mu_0} \vec{B}^2$

[48]



$$RI = -\mu_0 \pi a^2 H_0 \frac{d}{dt} \cos \omega t$$

$$I = \frac{\mu_0 \pi a^2 H_0 \omega}{R} \sin \omega t \quad //$$

$$dF = I \times B \, dl$$

$$M =$$

[49]

$$m n \left(\frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \operatorname{grad} \vec{v} \right) = - \operatorname{grad} p - e n \left(\vec{E} + \vec{v} \times \vec{B} \right)$$

$p = n k_B T$, $n \nabla \cdot \vec{v} + \frac{\partial n}{\partial t} = 0$

$\vec{j} = -e n \vec{v}$ $\nabla \cdot B = 0$ $\nabla \cdot D = \rho$

$\rho = e (n_0 - n)$ $\nabla \times E = - \frac{\partial B}{\partial t}$ $\nabla \times H = \frac{\partial D}{\partial t} + j$

---

wave $e^{i(\vec{k} \cdot \vec{x} - \omega t)}$ $\nabla = ik$ $\frac{\partial}{\partial t} = -i\omega$

$$m n \left(-i\omega \vec{v} + \vec{v} \cdot \underbrace{(i \vec{k} \cdot \vec{v})}_{i\omega} \right) = - i \vec{k} (n k_B T) - e n (\vec{E} + \vec{v} \times \vec{B})$$

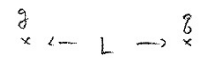
$$0 = i \vec{k} (k_B T) + e (\vec{E} + \vec{v} \times \vec{B})$$

$n i \vec{k} \cdot \vec{v} - i \omega n = 0 \quad \rightarrow \quad \vec{k} \cdot \vec{v} = \omega$

$\vec{k} \cdot \vec{B} = 0$ $i \vec{k} \cdot \vec{E} = \frac{e}{\epsilon_0} (n_0 - n)$

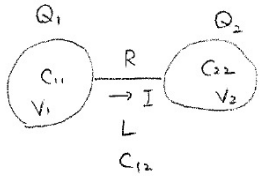
$\vec{k} \times \vec{E} = \omega \vec{B}$ $\frac{i \vec{k}}{\mu_0} \times \vec{B} = -i \epsilon_0 \omega \vec{E} - e n \vec{v}$

[50]



$B = \frac{\mu_0}{2\pi} \cdot \frac{q}{r^2} \hat{e}_r$ $\quad$ $E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2} \hat{e}_r$

[51]



$$Q_1 + Q_2 = Q$$

$$I = -\frac{dQ_1}{dt} = \frac{dQ_2}{dt}$$

$$RI + L\frac{dI}{dt} = \frac{Q_1}{C_{11}} - \frac{Q_2}{C_{22}}$$

$$I = \frac{dq}{dt}$$

$$q_1 = C_{11}V_1 + C_{12}V_2 \qquad q_1 - q = C_{11}V_1 + C_{12}V_2$$

$$q_2 = C_{12}V_1 + C_{22}V_2 \qquad q_2 + q = C_{12}V_1 + C_{22}V_2$$

$$V_1 - V_2 = RI + L\frac{dI}{dt}$$

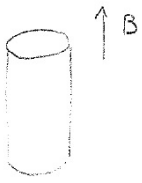
$$Q_1 = C_{11}V_1 + C_{12}V_2$$

$$Q_2 = C_{12}V_1 + C_{22}V_2$$

$$V_1 - V_2 = RI + L\frac{dI}{dt}$$

$$I =$$

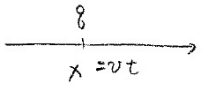
[5 2]



[53]

$$L = \frac{\epsilon_0}{2}\left(-\nabla\phi - \frac{\partial \vec{A}}{\partial t}\right)^2 - \frac{1}{2\mu_0}(\nabla \times \vec{A})^2 - \rho\phi + \vec{i}\cdot\vec{A}$$

[54]



$$E(x,t) = \frac{1}{4\pi\epsilon_0}$$